

Code Case – Pub/Sub using Function Apps

1. Implement Customer Pub/Sub

Prerequisite

1. Docker
2. Azure Function Core tools
3. .NET 9+

A basic .NET solution has been provided with bare stubs of the implementation. But you are allowed to explore other options as you see fit. It is expected this case should take no more than 5 hours.

A local dummy database with dummy database(customers.json) has been provided with methods for retrieving data from the database has also been provided. You are allowed to explore other options as database, but it is not a requirement.

A servicebus message factory, formatter, and repository has been provided.

A template file is provided with topics and subscriptions for the service bus.

This task revolves around creating two function apps. One publisher and one subscriber and using Azure Service Bus Emulator.

A docker-compose file has been provided to launch servicebus emulator.

1. Publisher requirement
 - a. Receive a JSON object which describes a customer
 - b. Send message through servicebus to the subscriber
2. Subscriber requirement
 - a. Receive message from the service bus
 - b. The subscriber should be able to handle both update and create customer
 - c. If an create/update fails in the subscriber then retry
 - d. If retries fails 3 times, convert the message to a deadletter

2. [Optional] Add automated test

Add automated test for the app functions.

1. Possible things to cover with tests
 - a. Test of publisher
 - i. Ensure the message sent to the servicebus has the correct format
 - ii. Trigger send message to servicebus
 - b. Test of subscriber
 - i. Can update an entry
 - ii. Deadletter'ing